

Optical Photothermal Infrared Spectroscopy for Microparticles Separated by Coffee Ring Effect



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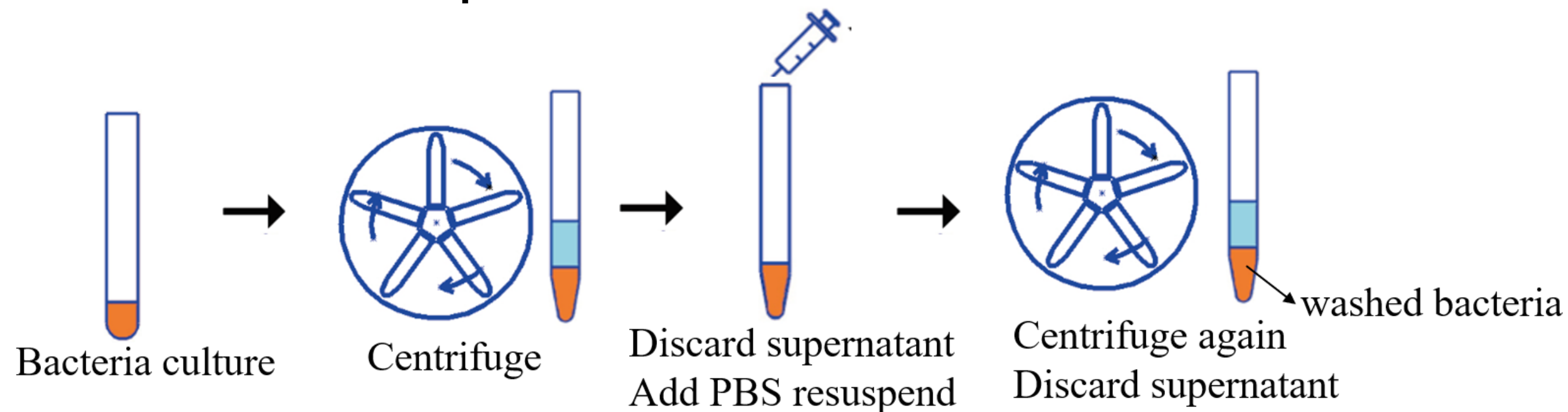
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Background

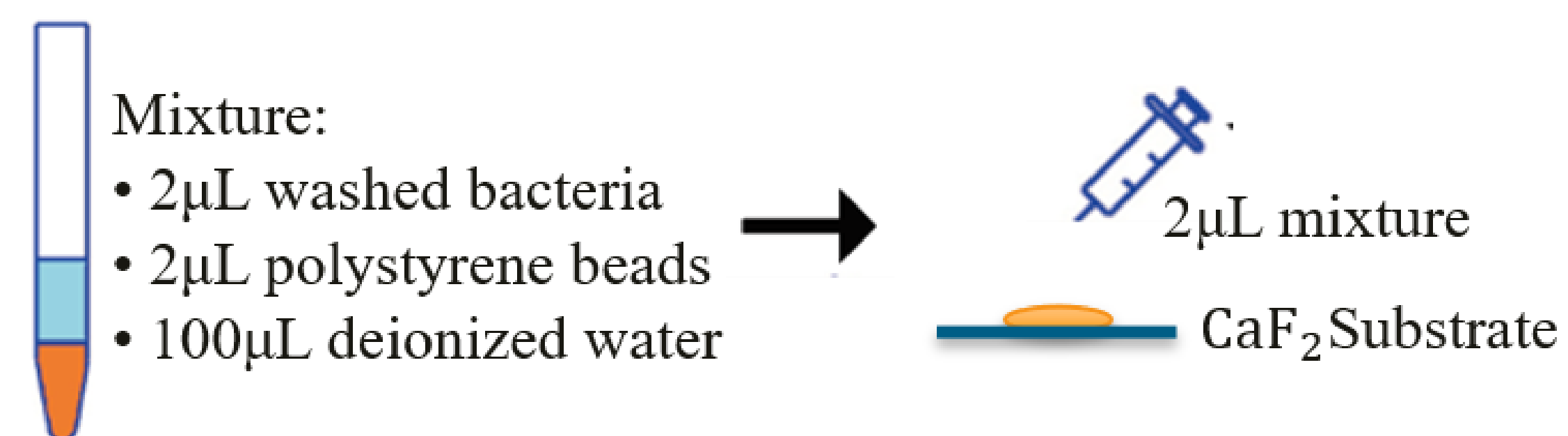
- Optical photothermal infrared (O-PTIR) spectroscopy uses a mid-infrared laser to cause a local thermal expansion. This expansion is detected using a visible beam, which measures changes in refractive index or scattered light.
- The coffee ring effect occurs when a droplet containing suspended particles evaporates, leaving a ring-like stain. Smaller particles are carried more easily to the boundary, while larger particles tend to remain more concentrated near the center.
- This work images a separation of two microparticles - a *Bacillus* bacteria (1 μm diameter) and polystyrene beads (3 μm diameter) via the coffee ring effect, where each is identified based on their distinctive infrared absorption characteristics using O-PTIR spectroscopy.

Methods

Bacteria Preparation:



Sample Preparation:



Experimental Setup:

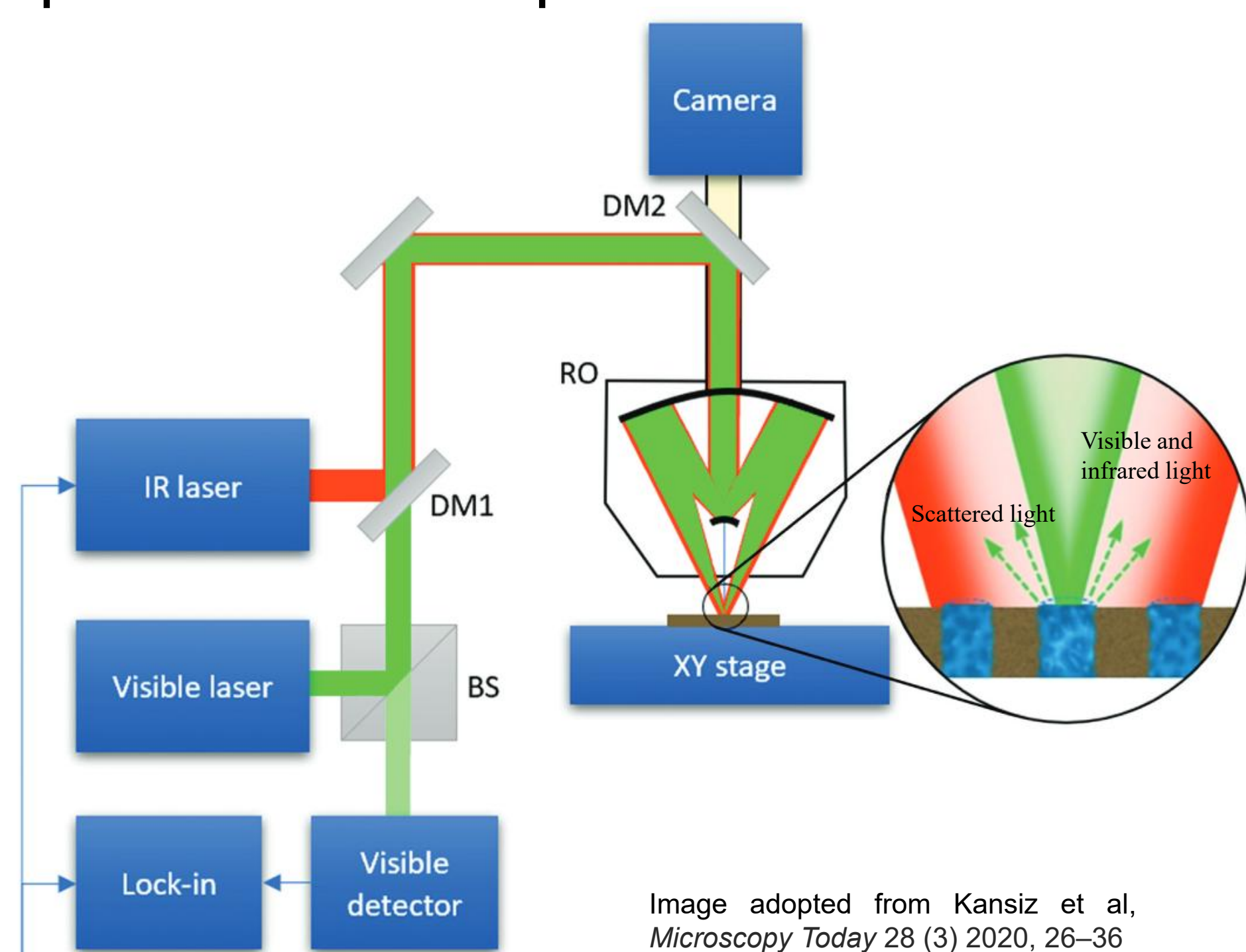
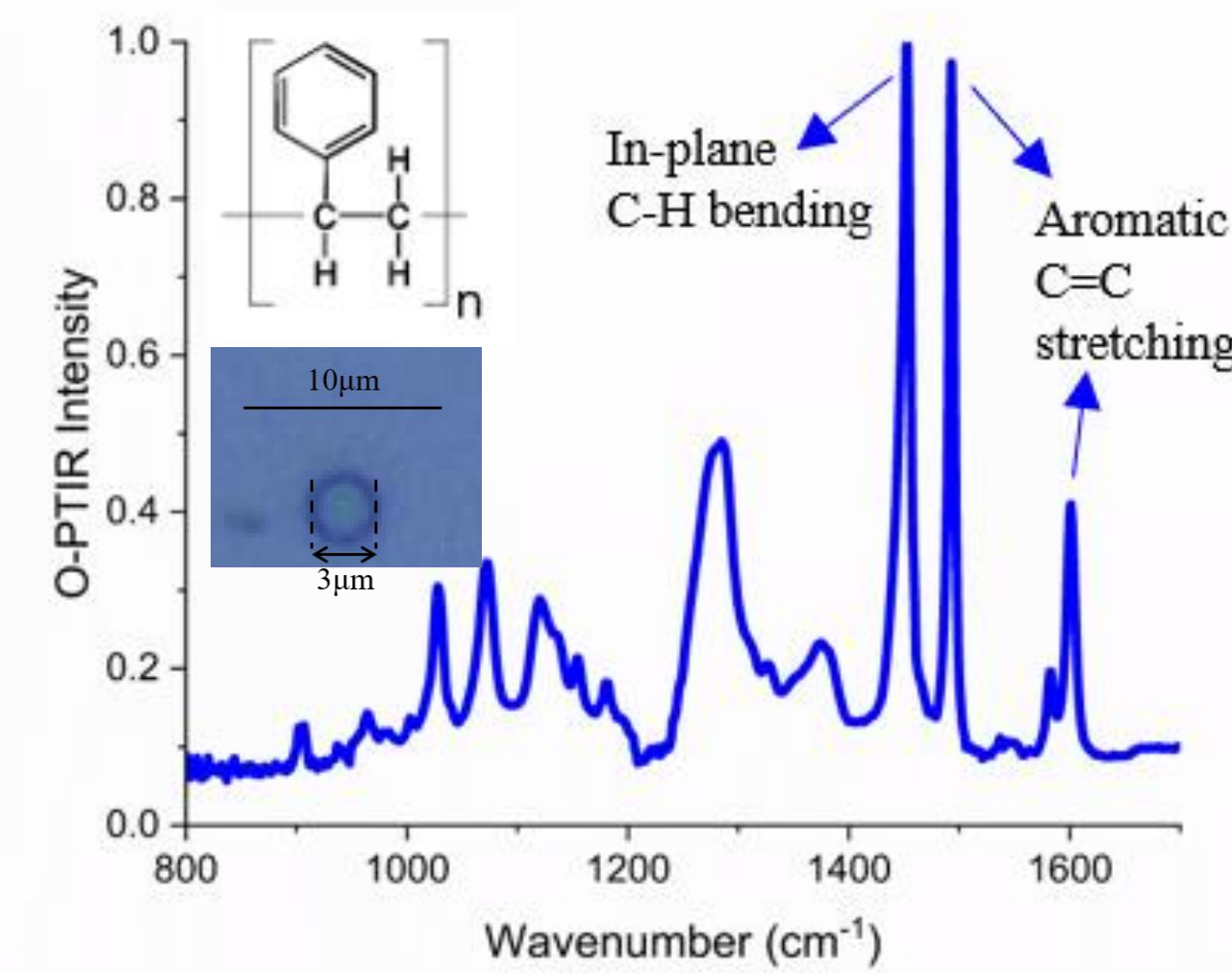
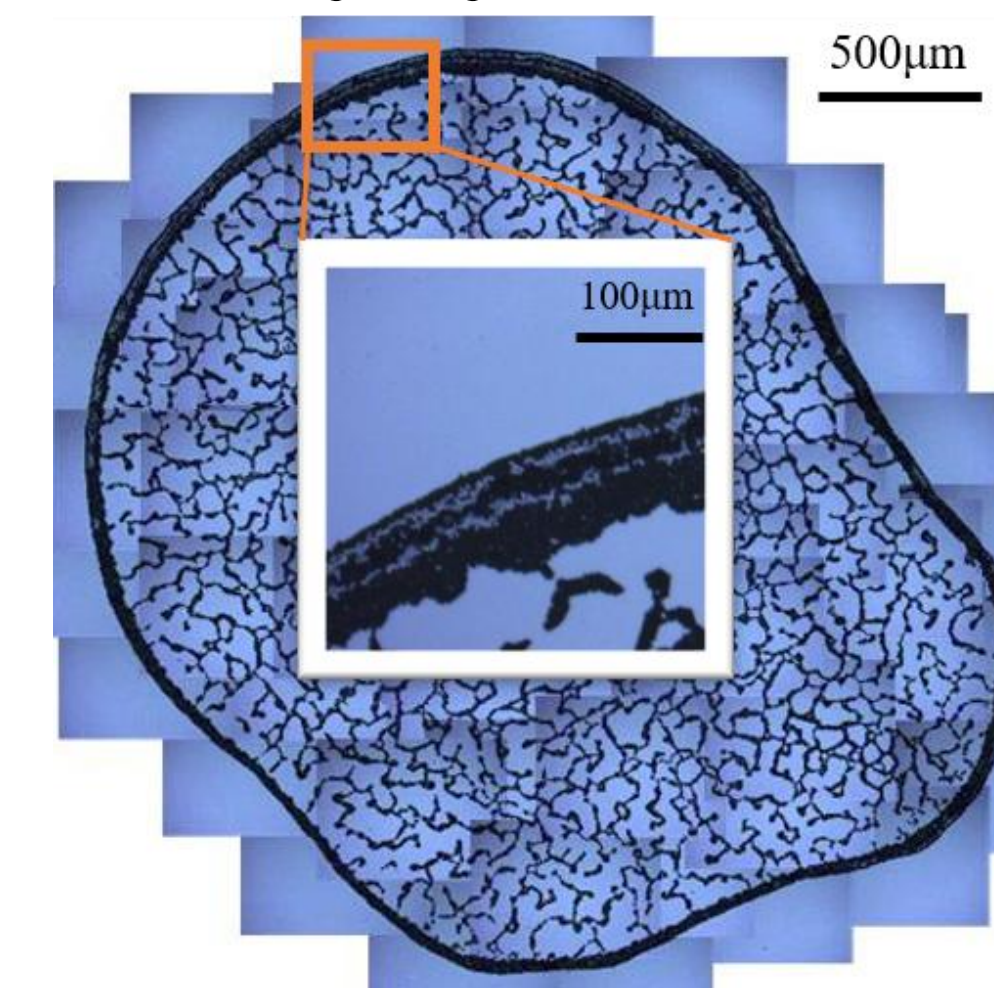


Image adopted from Kansiz et al, *Microscopy Today* 28 (3) 2020, 26–36

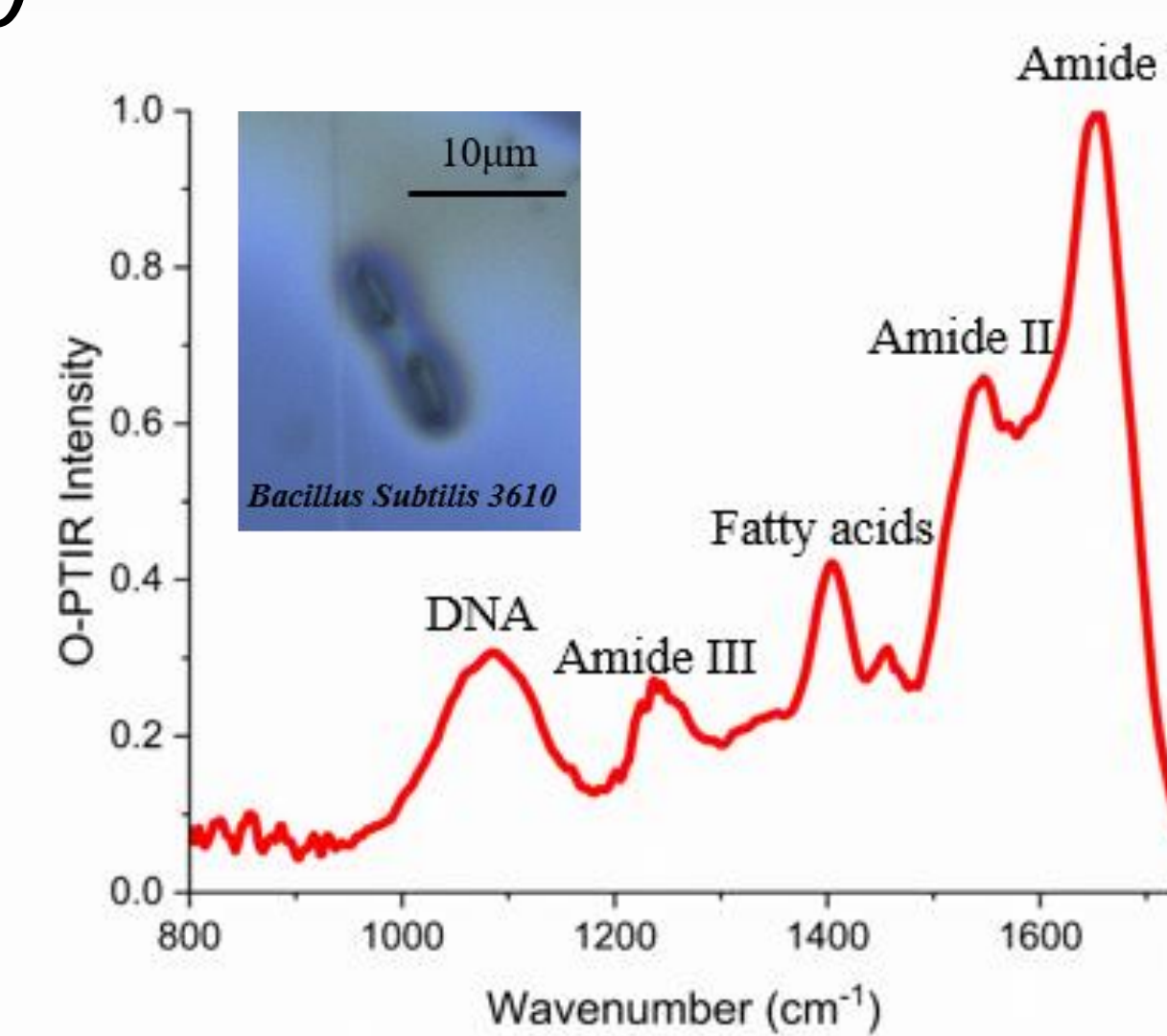
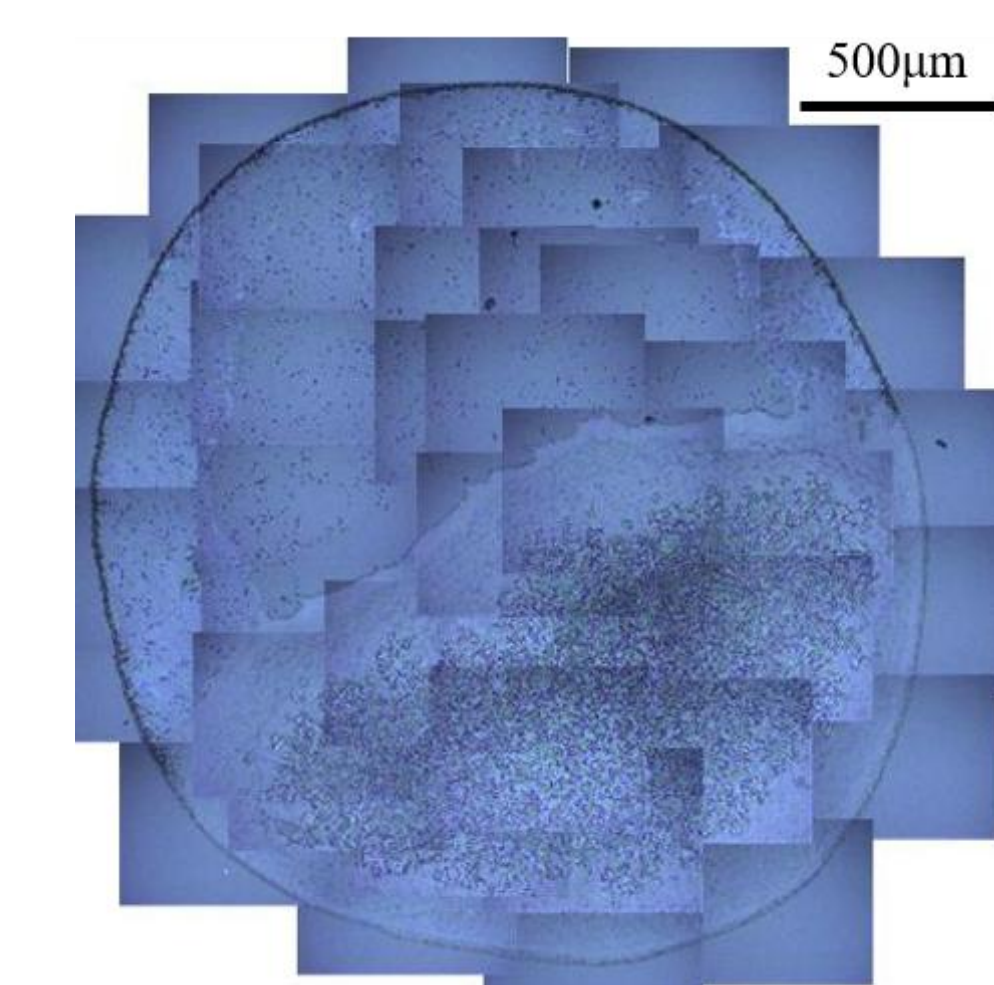
Results and Discussion

Individual Analysis

Polystyrene Beads

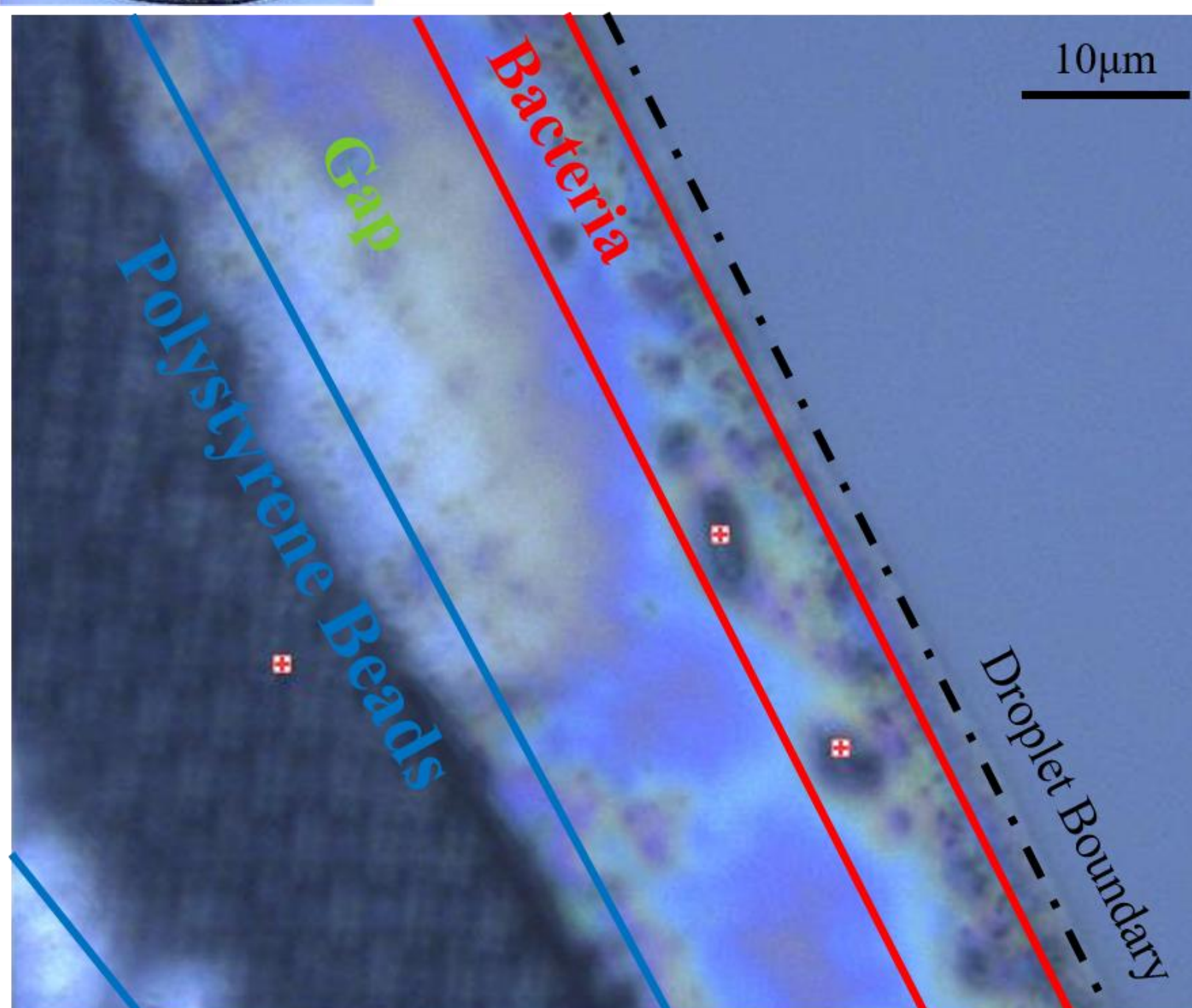
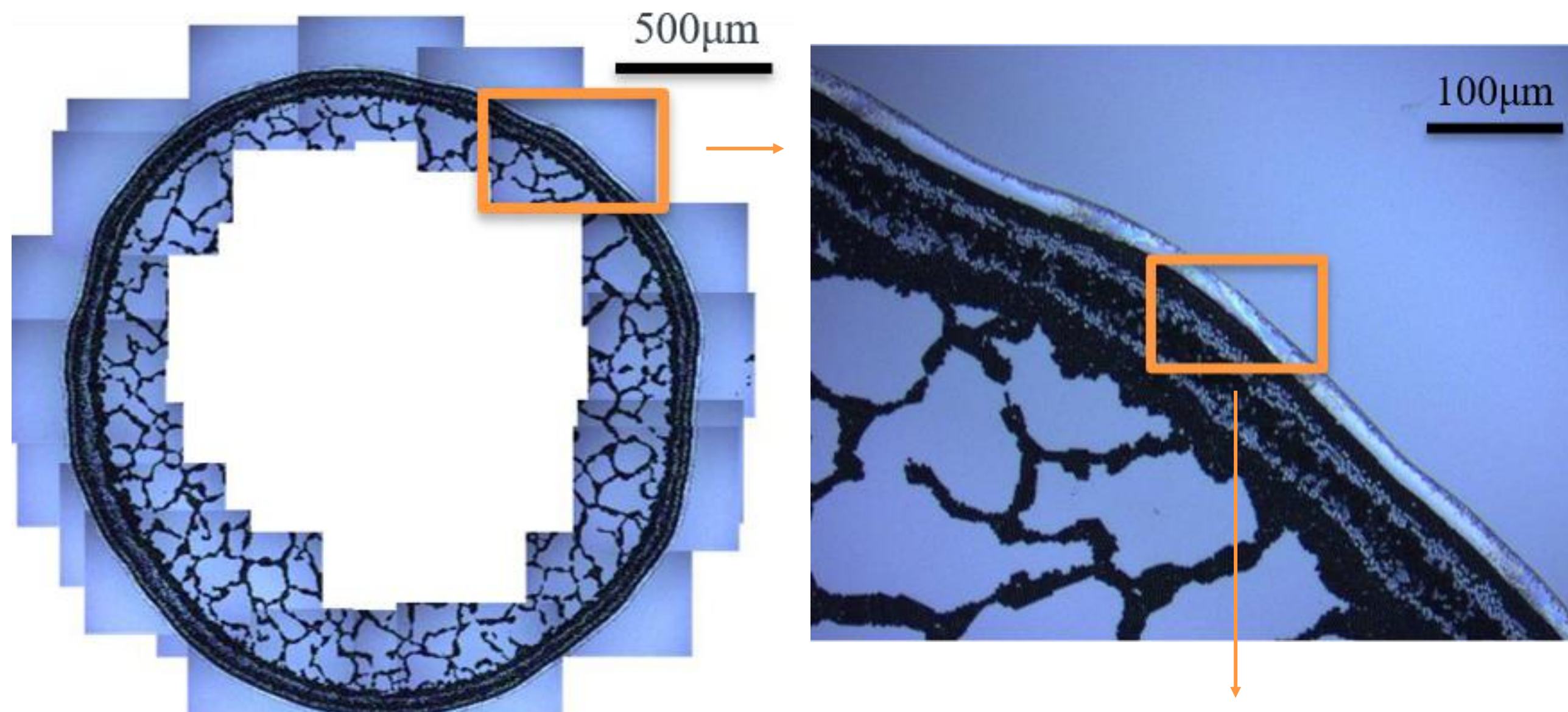


Bacillus subtilis 3610

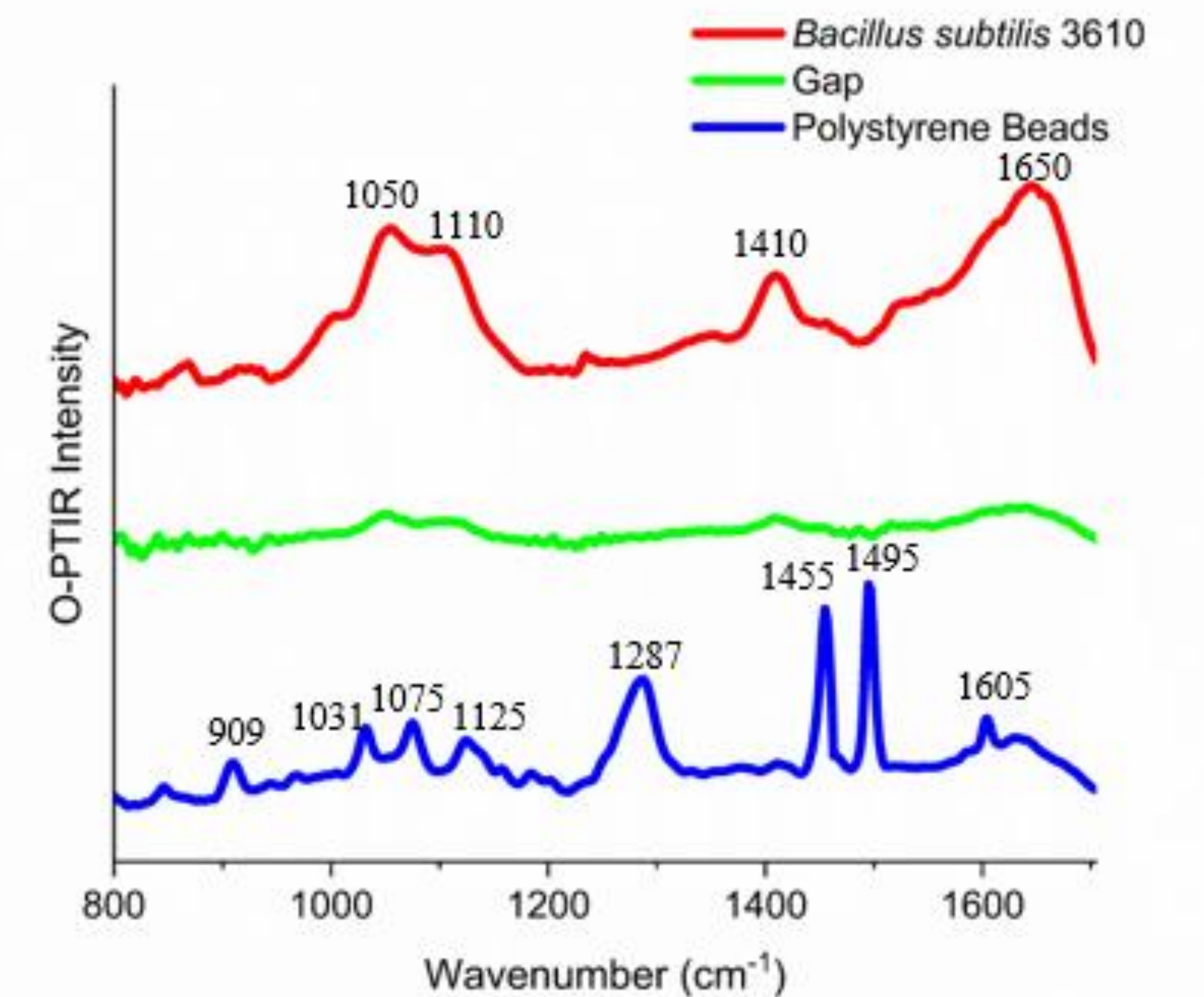


Mixture Analysis

- Ring structure formed after the 2 μL mixture droplet dried.



- O-PTIR measurement on bacteria, gap, and polystyrene beads regions.



- Color-coded consistently with the colors labeled in the mixture analysis region labels. The peaks are offset for clarity.
- The peaks of *Bacillus subtilis* bacteria and the polystyrene beads match their respective individual peaks.
- The spectrum of the gap region has no significant IR peaks, indicating that there are no bacteria or beads in this region.
- A larger gap distance can be achieved by a smaller contact angle between the droplet and the substrate.

Conclusions

- Samples were prepared by mixing bacteria and beads in deionized water, spotting the mixture on a substrate, and allowing it to dry. Optical photothermal infrared spectroscopy provided detailed spectral data from different regions of the dried droplet. The results showed bacteria at the outermost ring and polystyrene beads further inside, with a clear gap in between. Spectral analysis confirmed the presence of proteins and nucleic acids in the bacterial regions and aromatic rings in the polystyrene regions.
- This study suggests a simple method for microparticle separation, and when combined with infrared microscopy, able to quickly identify the particles.

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